I21-1707

DS-D

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Deep Learning Assignment

1. Introduction

Facial expression recognition and affective computing are important tasks in human–computer interaction. In this project, we implemented a **multi-task deep learning model** that simultaneously performs:

- **Expression classification** (8 categories), and
- **Valence–Arousal regression** (continuous values).

We evaluated two pretrained architectures:

- **ResNet50**
- **EfficientNetB0**

and compared their performance on a dataset of 3999 annotated images.

2. Dataset

- Dataset contained **3,999 samples**.
- Each image has four annotation types:
    - **Expression label** (exp.npy) → 8 classes
    - **Valence (val.npy)** and **Arousal (aro.npy)** → values in [-1, 1]
    - **Landmarks (lnd.npy)** (not used in this study).

Data split

- **Training**: 3,399 images (85%)
- **Validation**: 600 images (15%)
- Split was **stratified on expression labels** to maintain class balance.

3.1 Preprocessing

- Images resized to **224×224** pixels.
- Normalized to [0,1], with augmentations (flip, brightness, contrast).
- Multi-task label structure:
  - expression → one-hot encoded (8 classes)
  - va → 2D vector [valence, arousal]

3.2 Models

(a) ResNet50 (baseline)

- Pretrained on ImageNet.
- Global average pooling + dropout.
- Two heads:
  - **Classification**: Dense(8, softmax)
  - **Regression**: Dense(2, tanh)

(b) EfficientNetB0

- Pretrained on ImageNet.
- Stage 1: Base frozen → only dense heads trained.
- Stage 2: Fine-tuned last 20 layers (small learning rate).

3.3 Training setup

- **Optimizer**: Adam
- **Loss**:
  - Categorical Cross-Entropy (expression)
  - Mean Squared Error (VA regression)
- **Loss weights**: 1.0 (classification), 0.5 (regression)
- **Batch size**: 32
- **Epochs**: up to 15
- **Callbacks**: Early stopping, model checkpoint

4. Results

4.1 Expression Classification

Model	Validation Accuracy	Macro F1	Cohen's Kappa
ResNet50	**43.1%**	**0.43**	0.35
EfficientNetB0 (frozen)	12.5%	0.12	~0
EfficientNetB0 (fine-tuned)	31.0%	0.31	0.22

- ResNet50 achieved the best classification performance.
- EfficientNetB0 required fine-tuning to perform reasonably (~31%).
- Confusion matrices showed heavy **misclassification between similar emotions** (e.g., classes 0 vs 7, 2 vs 5).

4.2 Valence–Arousal Regression

ResNet50 (baseline):

- RMSE: [0.39 (val), 0.36 (aro)]
- CCC: [0.49, 0.39]
- SAGR: [0.75, 0.78]
- Pearson Corr: [0.57, 0.43]
- Accuracy: 0.43166666666666664
- F1-macro: 0.43200141666428826
- Cohen Kappa: 0.3504761904761905
- Classification Report:
- precision recall f1-score support

0 0.28 0.32 0.30 75

```
Classification Report:
              precision    recall  f1-score   support
```

```
           0       0.28      0.32      0.30        75
           1       0.63      0.60      0.62        75
           2       0.46      0.28      0.35        75
           3       0.46      0.39      0.42        75
           4       0.42      0.60      0.49        75
           5       0.45      0.41      0.43        75
           6       0.32      0.45      0.37        75
           7       0.59      0.40      0.48        75

    accuracy                           0.43       600
   macro avg       0.45      0.43      0.43       600
weighted avg       0.45      0.43      0.43       600

AUC-ROC (macro): 0.7986444444444445
AUC-PR (per-class): [np.float64(0.30111213708709794), np.float64(0.6795505428534863),
np.float64(0.4117180996490024), np.float64(0.4489567010589645),
np.float64(0.48808870987345154), np.float64(0.36385940762082425),
np.float64(0.3204151033992359), np.float64(0.46625040701717896)]
```

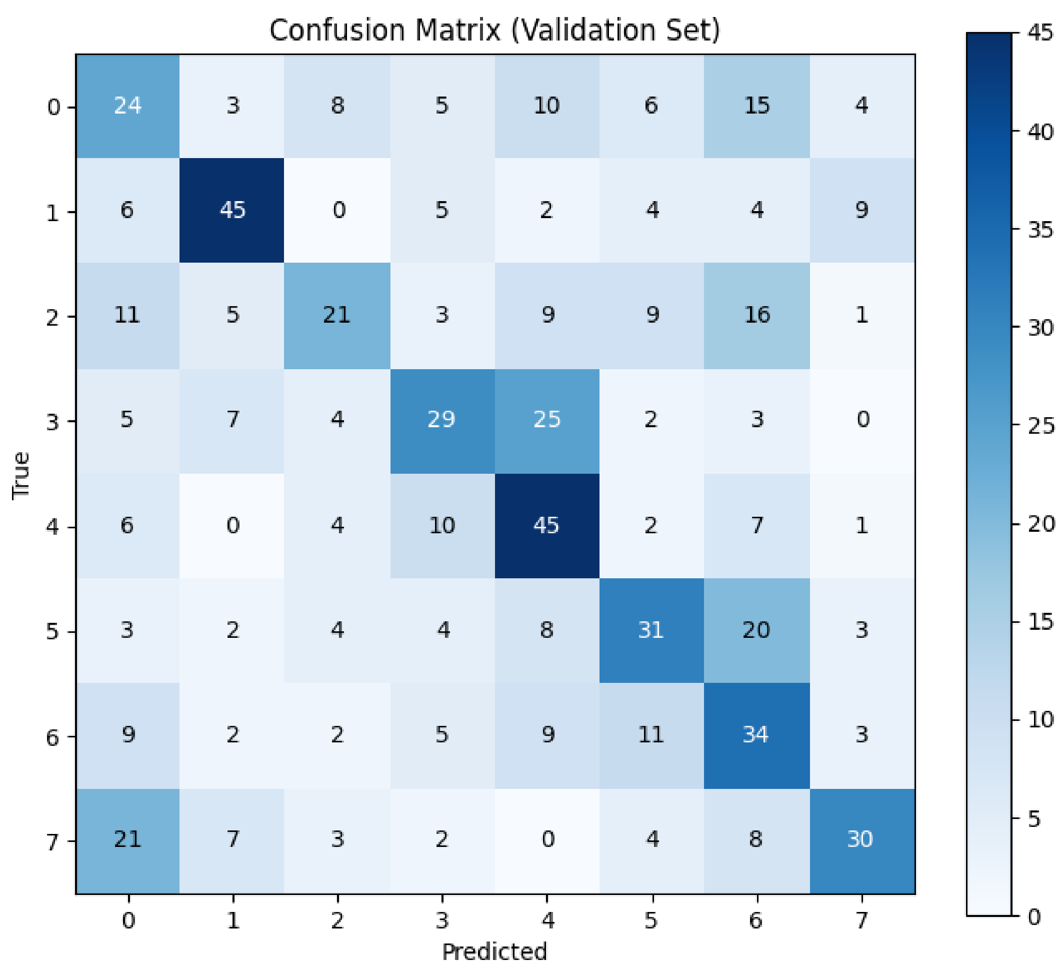
- AUC-PR (macro): 0.43499388856990523
- accuracy 0.43 600
- macro avg 0.45 0.43 0.43 600
- weighted avg 0.45 0.43 0.43 600
-
- AUC-ROC (macro): 0.7986444444444445
- AUC-PR (per-class): [np.float64(0.30111213708709794), np.float64(0.6795505428534863), np.float64(0.4117180996490024), np.float64(0.4489567010589645), np.float64(0.48808870987345154), np.float64(0.36385940762082425), np.float64(0.3204151033992359), np.float64(0.46625040701717896)]
- AUC-PR (macro): 0.43499388856990523

EfficientNetB0 (fine-tuned):

- MSE = 0.164
- MAE = 0.339
- R^2 = 0.083
- precision recall f1-score support
-
- 0 0.31 0.31 0.31 75
- 1 0.37 0.36 0.36 75
- 2 0.24 0.23 0.23 75

- 3 0.41 0.32 0.36 75
- 4 0.42 0.44 0.43 75
- 5 0.24 0.20 0.22 75
- 6 0.25 0.27 0.26 75
- 7 0.27 0.36 0.31 75
-
- accuracy 0.31 600
- macro avg 0.31 0.31 0.31 600
- weighted avg 0.31 0.31 0.31 600
- MSE: 0.16405870020389557
- MAE: 0.33944594860076904
- R2 Score: 0.08319631218910217

Confusion matrx:

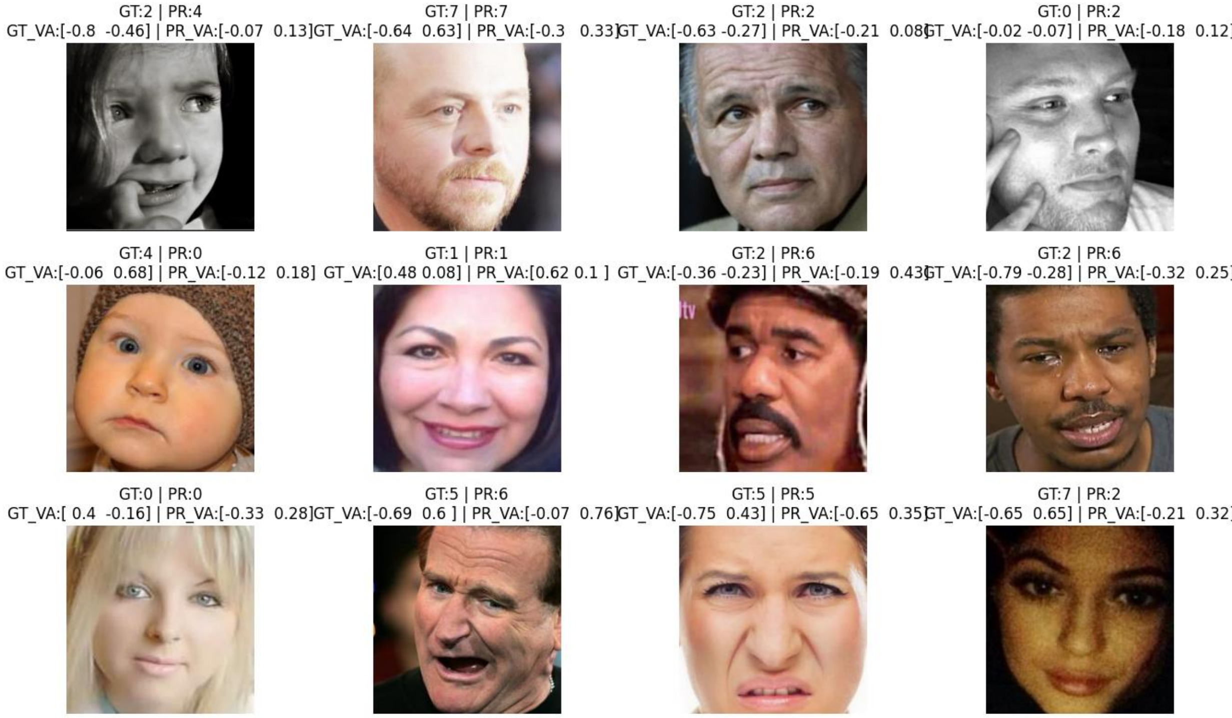


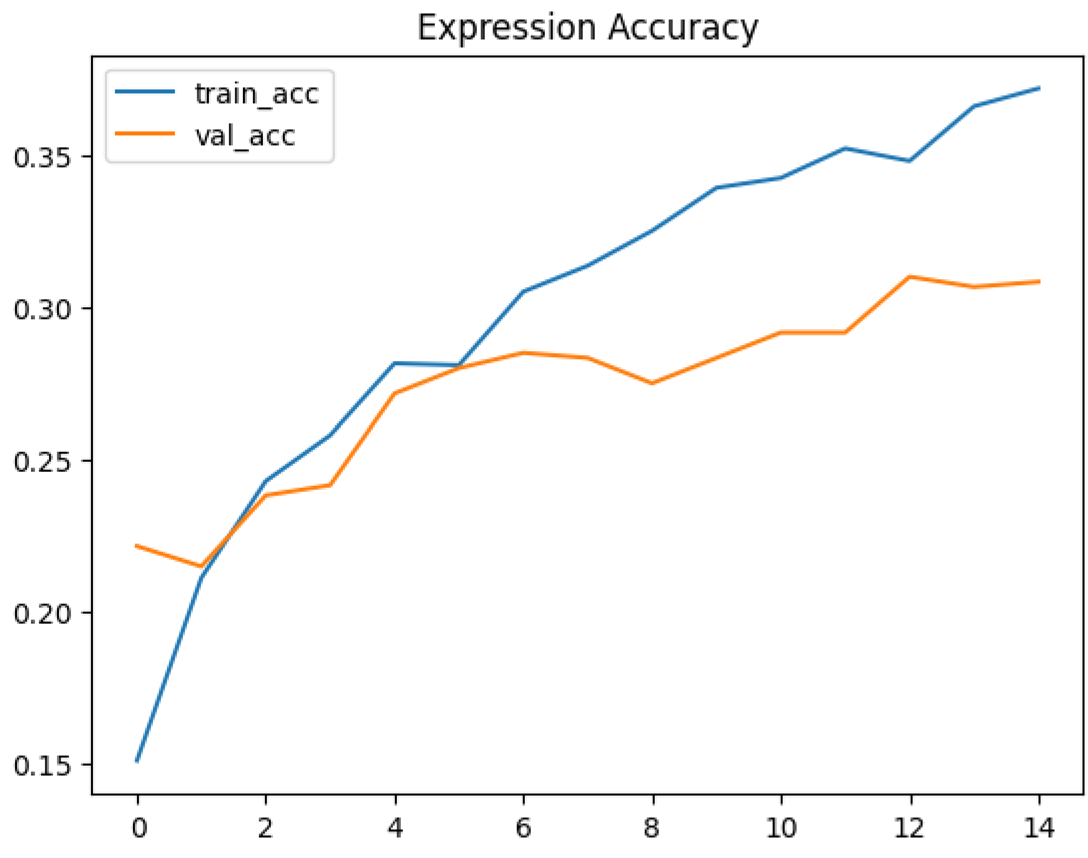
Interpretation:

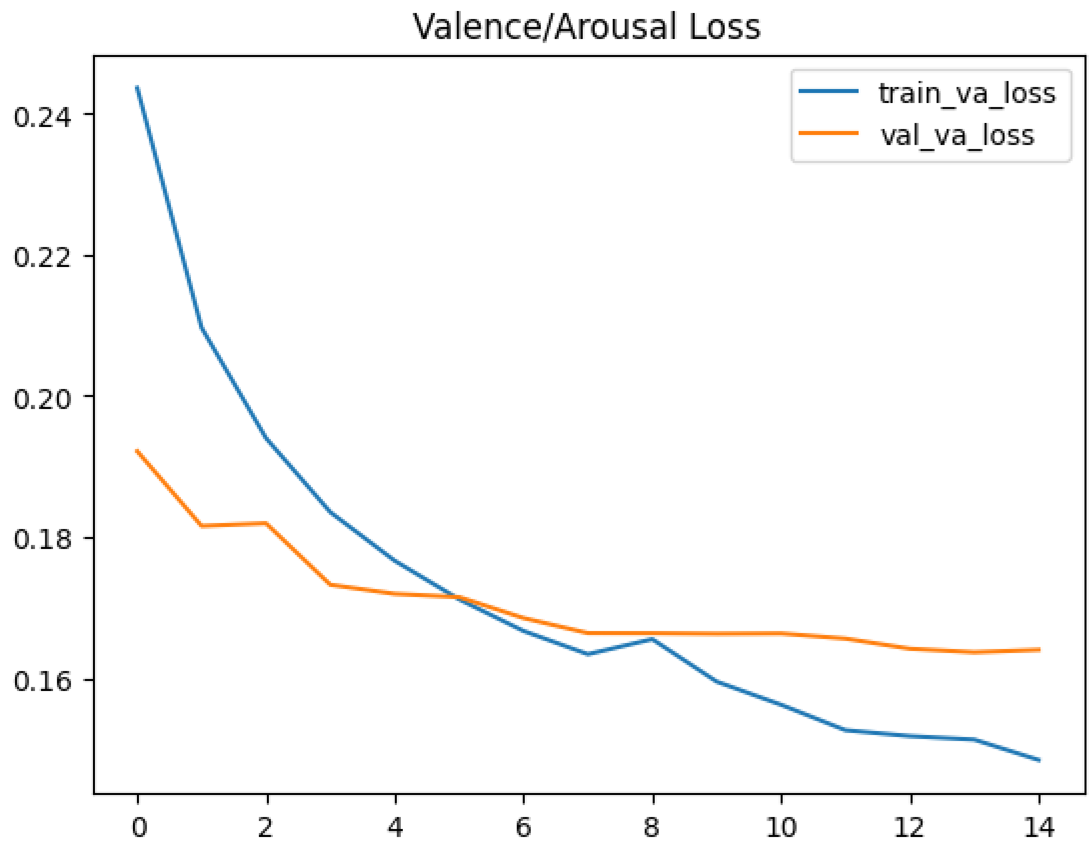
- Both models captured VA trends, but variance explained was low ($R^2 < 0.1$).
- ResNet50 outperformed EfficientNetB0 on CCC and correlation.

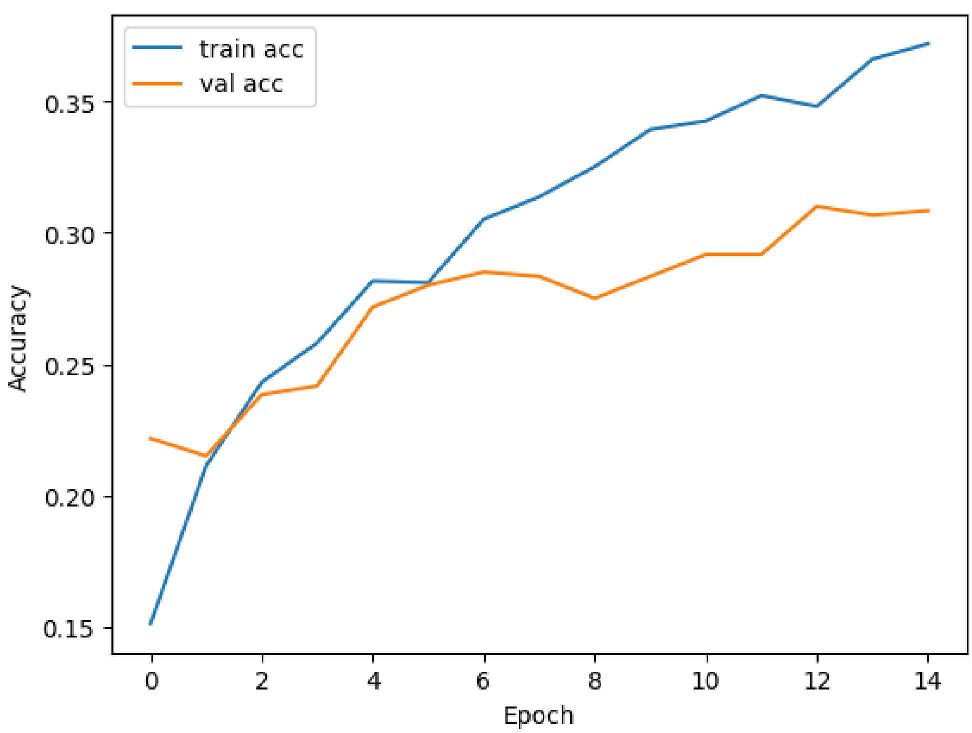
4.3 Training Curves

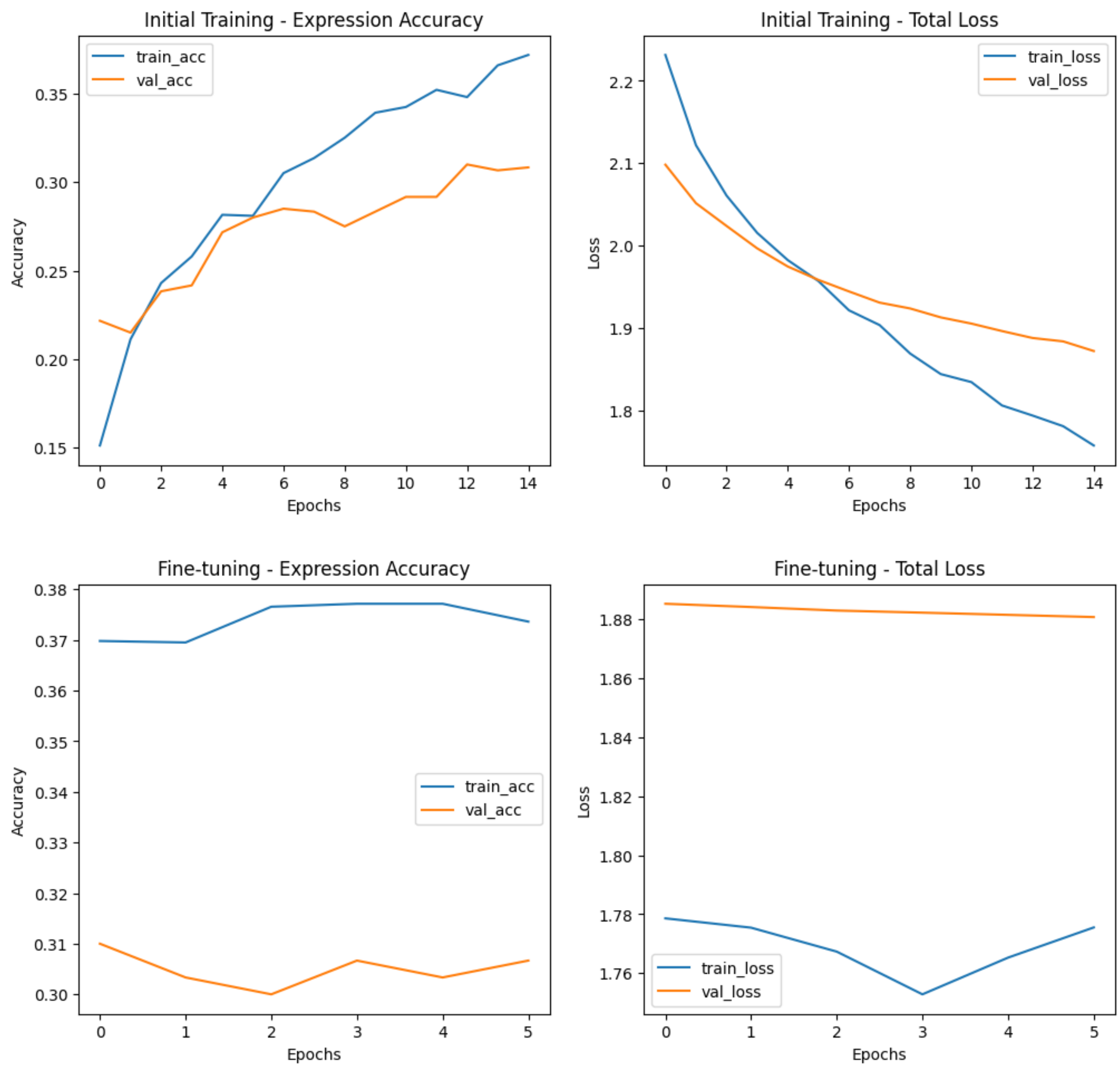
- **ResNet50**: Higher peak validation accuracy (~43%).
- **EfficientNetB0 (frozen)**: Flat at random guess (12.5%).
- **EfficientNetB0 (fine-tuned)**: Steady improvement to ~31% val acc.
- Loss curves showed no severe overfitting.


GT:2 | PR:4
GT_VA:[-0.8 -0.46] | PR_VA:[-0.07 0.13]
GT:7 | PR:7
GT_VA:[-0.64 0.63] | PR_VA:[-0.3   0.33]
GT:2 | PR:2
GT_VA:[-0.63 -0.27] | PR_VA:[-0.21 0.08]
GT:0 | PR:2
GT_VA:[-0.02 -0.07] | PR_VA:[-0.18 0.12]
GT:4 | PR:0
GT_VA:[-0.06 0.68] | PR_VA:[-0.12 0.18]
GT:1 | PR:1
GT_VA:[0.48 0.08] | PR_VA:[0.62 0.1]
GT:2 | PR:6
GT_VA:[-0.36 -0.23] | PR_VA:[-0.19 0.43]
GT:2 | PR:6
GT_VA:[-0.79 -0.28] | PR_VA:[-0.32 0.25]
GT:0 | PR:0
GT_VA:[0.4 -0.16] | PR_VA:[-0.33 0.28]
GT:5 | PR:6
GT_VA:[-0.69 0.6] | PR_VA:[-0.07 0.76]
GT:5 | PR:5
GT_VA:[-0.75 0.43] | PR_VA:[-0.65 0.35]
GT:7 | PR:2
GT_VA:[-0.65 0.65] | PR_VA:[-0.21 0.32]


Expression Accuracy
train_acc
val_acc
0.35
0.30
0.25
0.20
0.15
0
2
4
6
8
10
12
14


Valence/Arousal Loss
train_va_loss
val_va_loss
0.24
0.22
0.20
0.18
0.16
0
2
4
6
8
10
12
14


train acc
val acc
Accuracy
0.35
0.30
0.25
0.20
0.15
0
2
4
6
8
10
12
14
Epoch



5. Discussion

- **ResNet50** transferred better to this dataset, achieving stronger classification and regression performance.
- **EfficientNetB0** struggled without fine-tuning; after unfreezing, accuracy improved but still lagged behind.
- Likely reasons:
    - Dataset size (~4k) is small for EfficientNet's high capacity.
    - Emotion recognition is inherently ambiguous, leading to confusion across classes.

- **Regression results** show both models learn valence–arousal trends but struggle to capture variance.

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6. Conclusion

- **Best overall model**: ResNet50 (43% accuracy, better VA metrics).
- EfficientNetB0 showed improvement with fine-tuning but underperformed compared to ResNet50..